H1 — Geopolitical Risk as a Cost-Push Shock

Updated specification: ADL + LP, unified 1990–2025 sample

Hypothesis (H1)

A positive GPR shock raises consumer prices (inflation ↑) and lowers industrial production (output ↓) — the cost-push signature.

1. Data

1.1 Variables

Table 1 lists all variables used in the analysis. The baseline specification uses only the shock and outcome variables; robustness controls are introduced in Section 3.

Table 1. Variable Definitions

Variable	Code	Source	Freq.	Start	Unit	Transformation
Shock variables						
GPR (headline)	—	C&I[1]	M	1900	Index	$100\log(100 \cdot \text{GPR})$
GPT (threats)	—	C&I	M	1900	Index	$100\log(100 \cdot \text{GPT})$
GPA (acts)	—	C&I	M	1900	Index	$100\log(100 \cdot \text{GPA})$
Outcome variables						
Ind. Production	INDPRO	FRED	M	1919	Idx (2017=100)	$100\log(\text{IP})$ & $100\Delta\log(\text{IP})$
CPI (headline)	CPIAUCSL	FRED	M	1947	Idx (82–84=100)	$100\log(\text{CPI})$ & $100\Delta\log(\text{CPI})$
Robustness controls						
Unemployment	UNRATE	FRED	M	1948	Percent	Level
VIX	VIXCLS	FRED	D→M	1990	Index	$100\log(\text{VIX})$

Shock normalisation. The three GPR measures enter as $100\log(100 \cdot x_t)$. The inner factor of 100 is an affine rescaling inherited from the team's preliminary code (Baek–Lee convention); it shifts the constant but leaves all impulse-response shapes and confidence bands invariant. The outer $\times 100$ expresses shock innovations in log-points so that IRF magnitudes are directly comparable across specifications. All IRFs are reported per *one standard deviation* of the shock, so the specific affine rescaling does not affect economic interpretation.

1.2 Stationarity

Unit root properties are verified with Augmented Dickey–Fuller (ADF) regressions, $\Delta x_t = a + \rho\, x_{t-1} + \sum_{j=1}^{12} \phi_j \Delta x_{t-j} + e_t$, with a constant and 12 monthly augmentation lags, on the full available sample prior to the 1990:01 trim (`manual_adf.m`, no Econometrics Toolbox dependency). MacKinnon (1996) 1% critical value with constant and $T \approx 500$ is -3.44.

Table 2. ADF Unit Root Tests — t-statistics on lagged-level coefficient ρ

Variable	Level t	1st-diff t	1% CV	Order
log(GPR)	−3.155	−8.606	−3.440	$I(1)$
log(GPT)	−3.012	−8.696	−3.440	$I(1)$
log(GPA)	−2.966	−8.046	−3.440	$I(1)$
log(IP)	−2.673	−5.104	−3.440	$I(1)$
log(CPI)	0.465	−4.975	−3.440	$I(1)$
log(WTI)	−1.566	−6.271	−3.440	$I(1)$
log(VIX)	−3.288	−7.517	−3.440	$I(1)$
UNRATE	−2.512	−6.117	−3.440	$I(1)$
FFR	−2.838	−4.805	−3.440	$I(1)$

Reading the table. At the strict 1% level none of the nine level series reject the unit-root null, whereas every first difference rejects decisively. All analysis variables are therefore classified $I(1)$. Note that at looser thresholds the picture shifts somewhat: log(GPR), log(GPT), log(GPA), and log(VIX) reject at the 5% level (−2.87), and log(IP), UNRATE, and FFR reject at the 10% level (−2.57), so these series are best described as borderline / highly persistent rather than pure random walks. This is fully consistent with the literature (GPR is persistent but mean-reverting; VIX exhibits near-unit-root AR(1) dynamics; UNRATE is I(1) at monthly frequency over this sample).

Implication for the design. Unit-root status guides the choice of LHS transformation, not the validity of inference:

- **LP.** The local projection uses the *cumulative log change* $y_{t+h} - y_{t-1}$ as dependent variable, passing log(IP) or log(CPI) directly and letting `ir_jorda.m` form the cumulative difference internally (`level=0` branch). Because the LHS is itself a difference, LP is robust to $I(1)$ outcomes without requiring stationarity of the level.
- **ADL.** The Baek–Lee distributed-lag regression uses $\Delta\log(\text{IP})$ and $\Delta\log(\text{CPI})$ as LHS, which are $I(0)$ by the ADF results above; the routine cumulates the growth-rate IRF to a level response internally.
- **log(VIX) as a control.** The R2 specification enters log(VIX) in level. Although the level ADF is borderline ($t = -3.29$, rejects at 5% but not at 1%), the level is the economically correct transformation because we want to absorb persistent high-uncertainty *regimes*; first-differencing VIX would destroy that regime information.

Both methodologies therefore report impulse responses in *log-level* units, and the shock–outcome pairs are constructed so that each regression's LHS is stationary or robust to $I(1)$ regressors by construction.

1.3 Sample Period

Table 3. Sample Design

Parameter	Value / Rationale
Start	1990:01
End	2025:12
T	$\approx$ 432 months
Rationale	VIX (VIXCLS) only begins 1990:01. To ensure all three specifications (Baseline, R1, R2) are estimated on an *identical* time window, the entire sample is trimmed to 1990:01+.

Design choice. The previous draft used 1985:01 as the start and dropped the first five years only for Robustness 2 (which adds VIX). That made R2 non-comparable to Baseline/R1 in sample size. The updated design fixes T = 1990:01–2025:12 across all specifications, yielding clean apples-to-apples comparison of $\hat{\beta}_h$ across the three robustness layers. The trade-off — losing 60 observations at the start — is small relative to the identification benefit.

2. Model Specification

We estimate the GPR impulse response with *two* methodologies in parallel: (i) an autoregressive distributed lag (ADL) model following Baek & Lee (2022), and (ii) local projections (LP) following Jordà (2005). The two methods make different assumptions about the transmission dynamics; agreement between them strengthens the identification argument.

2.1 ADL (Baek & Lee 2022)

For each shock–outcome pair we estimate, via the IC-selected ADL(p, q),

$$\Delta y_t = c + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + \sum_{j=0}^{q} \theta_j x_{t-j} + \varepsilon_t \tag{1}$$

where Δy_t is $\Delta \log(\text{IP})_t$ or $\Delta \log(\text{CPI})_t$ (i.e., inflation), x_t is the shock, and IRFs are recovered by converting the ADL to its MA(∞) representation and cumulating to level form. Point estimates use OLS; confidence bands use Monte Carlo simulation ($N_B = 1000$). The endogenous lag length p is chosen by BIC from candidates $\{1, \ldots, 12\}$; $q = H = 36$.

2.2 LP (Jordà 2005)

For each horizon $h = 0, 1, \ldots, H$ we estimate by OLS:

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{cumulative log change}} = \alpha_h + \beta_h \cdot \text{shock}_t + \sum_{\ell=1}^{L} \gamma_{h,\ell}\, \Delta y_{t-\ell} + \sum_{\ell=1}^{L} \delta_{h,\ell}\, \text{shock}_{t-\ell} + \varepsilon_{t+h} \tag{2}$$

where y is log(IP) or log(CPI) (both multiplied by 100 so IRF units are log-points $\times 100 \approx$ percent). The sequence $\{\beta_0, \beta_1, \ldots, \beta_H\}$ is the level impulse response function. Inference uses Newey–West HAC with bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$ (Stock–Watson rule of thumb).

Table 4. LP Specification — Component Details

Component	Notation	Description
Dependent variable	$y_{t+h} - y_{t-1}$	Cumulative log change in the outcome from $t-1$ to $t+h$. Using cumulative differences (rather than h-step-ahead levels) makes the LP robust to $I(1)$ outcomes; the implementation passes log(IP) or log(CPI) directly and the routine forms $y_{t+h} - y_{t-1}$ internally.
Shock	shock_t	$100 \log(100 \cdot \text{GPR}_t)$, $100 \log(100 \cdot \text{GPT}_t)$, or $100 \log(100 \cdot \text{GPA}_t)$. Three parallel regressions, one per shock variant.
Own lags	$\Delta y_{t-\ell}$	Differenced outcome lags absorb short-run persistence and prevent spurious long-horizon correlations.
Own lags of shock	$\text{shock}_{t-\ell}$	Absorb persistence in GPR and control for dynamic feedback.
Trend	—	Linear trend included (`trend=1`).
β_h (key)	—	Impulse response at horizon h: the cumulative log-point response of the outcome to a unit shock. Reported below per *one standard deviation* of the shock.

2.3 Why No Additional Controls in the Baseline?

The baseline deliberately excludes oil prices, the federal funds rate, and credit spreads. These are *mediators* (transmission channels), not confounders. Controlling for a mediator removes part of the causal effect we are trying to measure, attenuating β_h.

Table 5. Mediator vs. Confounder Classification

Variable	Type	Causal path / Reason excluded
Oil (WTI)	Mediator	GPR $\to$ oil $\to$ CPI (dominant supply-side channel)
FFR	Mediator	GPR $\to$ π $\to$ Fed $\to$ FFR (monetary-policy offset)
Credit spread	Mediator	GPR $\to$ risk premia $\to$ spreads
UNRATE	Pot. confounder	Included in Robustness 1
VIX	Pot. confounder	Included in Robustness 2

Identification assumption (Caldara & Iacoviello 2022). GPR is driven by geopolitical events (wars, terrorism, political crises) exogenous to the US business cycle within the month. Under this assumption, own lags are sufficient for consistent estimation of β_h.

3. Robustness Controls

Two robustness specifications progressively add potential confounders. Equation (2) becomes

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \cdot \text{shock}_t + \sum_{\ell=1}^{L} \gamma_{h,\ell}\, \Delta y_{t-\ell} + \sum_{\ell=1}^{L} \delta_{h,\ell}\, \text{shock}_{t-\ell} + \sum_{\ell=1}^{L} \boldsymbol{\phi}'_{h,\ell} \mathbf{W}_{t-\ell} + \varepsilon_{t+h} \tag{3}$$

where $\mathbf{W}$ contains the robustness controls (analogous augmentation for the ADL).

Table 6. Robustness Specifications

Spec.	W	Justification
Baseline	(none)	Pure total effect. Assumes GPR exogeneity (C&I 2022).
Robustness 1	UNRATE	Unemployment proxies the business-cycle state: a recession may (a) raise GPR through political instability and (b) independently affect inflation via the Phillips curve. UNRATE is *not* a mediator because GPR's primary transmission runs through supply-side channels (oil, commodities), not through unemployment.
Robustness 2	UNRATE, log(VIX)	Global financial uncertainty (VIX) could simultaneously (a) raise media attention to geopolitical events, inflating the GPR measurement, and (b) depress investment/consumption. C&I (2022, Table 3) show GPR and VIX are moderately correlated ($\rho \approx 0.3$–0.4) but capture distinct constructs. VIX enters in **log-level** (stationary), absorbing persistent uncertainty regimes (2008, 2020); first-differencing would discard level information.

Interpretation. If β_h remains stable across all three specifications, the GPR exogeneity assumption is supported. Substantial shifts when adding UNRATE or VIX would signal potential confounding.

4. Lag Selection and Parameters

4.1 LP Lag Length: $L = 12$

Unlike the ADL, where the endogenous lag order is IC-selected, the LP lag length is a *user-imposed* control. We set $L = 12$ based on four independent criteria:

Table 7. Rationale for $L = 12$ (LP specification)

Criterion	Detail
Literature precedent	Caldara & Iacoviello (2022, AER) use 12 lags in their monthly VAR for GPR impulse responses. Matching their lag length ensures direct comparability of our LP IRFs to their VAR IRFs.
Seasonality absorption	Twelve monthly lags span one full year, absorbing seasonal autocorrelation in CPI and IP without requiring explicit seasonal dummies. Shorter LP lag lengths can leave residual seasonal serial correlation.
Persistence of the shock	GPR is highly persistent (see Figure 4: ACF remains above 0.3 past lag 12 for all three variants). Omitting long shock lags would bias β_h upward because past GPR predicts both current GPR and future outcomes.
Degrees of freedom	$T \approx 432$, $L = 12$, two lagged series (shock + outcome): $2 \times 12 + 2 = 26$ RHS parameters. At the longest horizon $h = 36$, $T_{\text{eff}} \approx 384$ — well above any degree-of-freedom concern.

LP $\neq$ VAR for lag selection. Plagborg-Møller & Wolf (2021, *Econometrica*) show that LP and VAR estimate the same impulse responses asymptotically, but the role of lags differs fundamentally. In a VAR, the lag order is a model-selection parameter that *determines the dynamics*; in an LP, lags are *controls* for serial correlation — they prevent spurious long-horizon correlations but do not define the dynamic structure. For LP, erring on the side of *more* lags is safer than too few: underfitting risk is low, while insufficient lags risk residual serial correlation and biased HAC inference. Hence $L = 12$ is conservatively safe rather than "optimal" in an information-criterion sense.

4.2 ADL Lag Length: BIC-Selected

The ADL endogenous lag order p is selected by BIC from candidates $\{1, 2, \ldots, 12\}$ separately for each (shock, outcome) pair. This differs from the LP convention because in the ADL the lag structure *is* the dynamic model, so parsimony matters. The shock lag count q is fixed at $q = H = 36$ so that the MA(∞) inversion is well-defined out to the maximum horizon. The selected p values for each specification are printed by the run script.

4.3 Estimation Parameters

Table 8. Estimation Parameters

Parameter	Value / Rationale
H (max horizon)	36 months
L (LP lag length)	12 — matches C&I (2022, AER); absorbs one full year of seasonality
p (ADL endogenous lags)	BIC-selected from $\{1, \ldots, 12\}$
q (ADL shock lags)	36 ($= H$)
α (CI level)	0.10 — 90% confidence bands (standard in macro LP literature)
LP trend	Linear
LP SE	Newey–West HAC, bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$
ADL SE	Monte Carlo, $N_B = 1000$
Critical value	$z_{0.05} = 1.645$ (normal approximation, 90% two-sided)

5. Results

5.1 Descriptive Analysis

Figure 1 plots the three GPR indices. Major geopolitical events (Gulf War 1990–91, 9/11 attacks, Iraq War 2003, Russia–Ukraine conflict 2022) produce visible spikes. GPA (acts) exhibits sharper, shorter spikes than GPT (threats), consistent with the distinction between realised events and anticipatory news coverage.

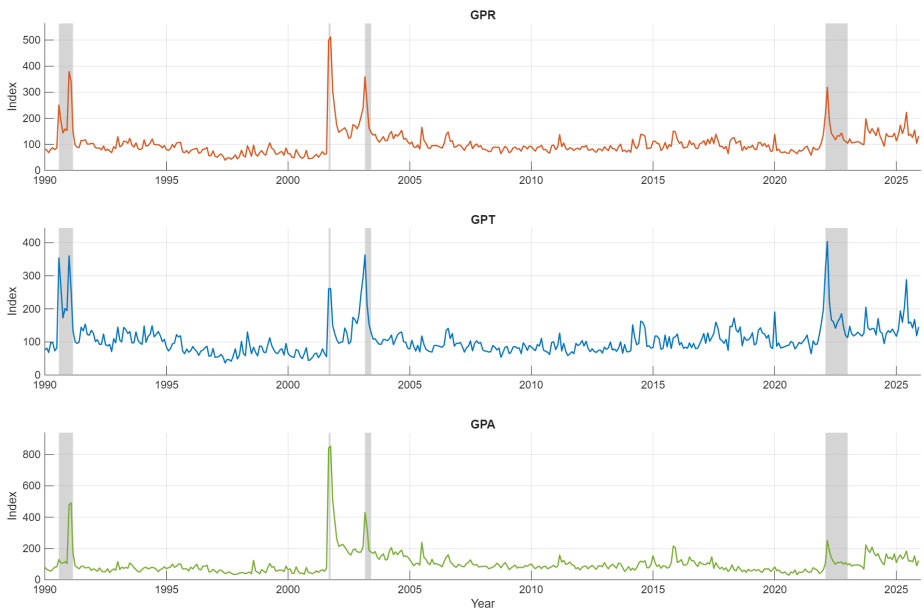

Figure 1. GPR / GPT / GPA indices, 1990–2025. Shaded bands: Gulf War, 9/11, Iraq War, Russia–Ukraine conflict.

Figure 3 presents the pairwise correlation matrix of analysis variables. Key observations: (i) the three GPR measures are highly correlated with each other ($\rho = 0.57$–0.89), confirming they capture related but non-identical constructs; (ii) contemporaneous correlations with macro variables are generally modest, suggesting that transmission is dynamic (lag-dependent) rather than contemporaneous; (iii) inflation and oil-price growth are positively correlated, as expected.

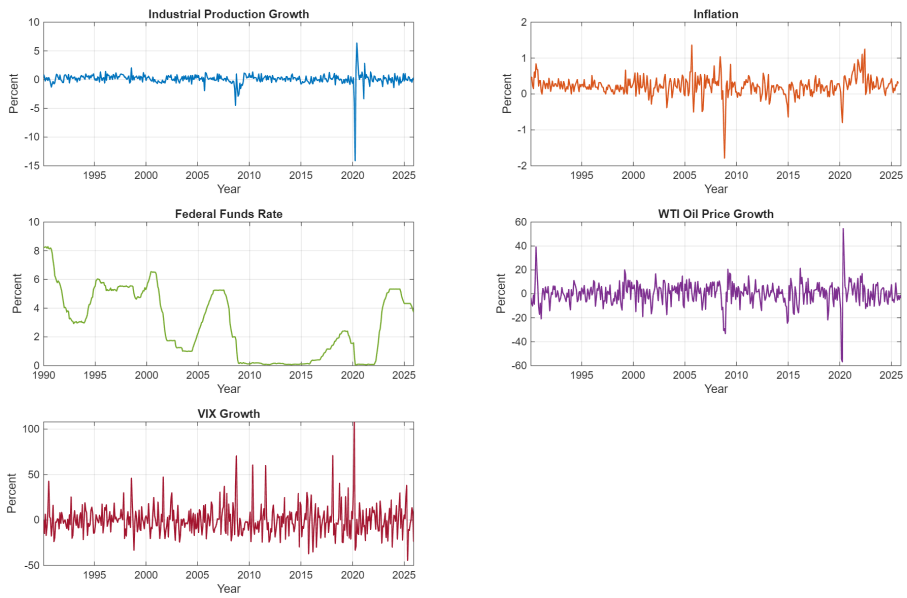

Figure 2. Macro variables: IP growth, inflation, FFR, WTI growth, VIX growth (1990–2025).

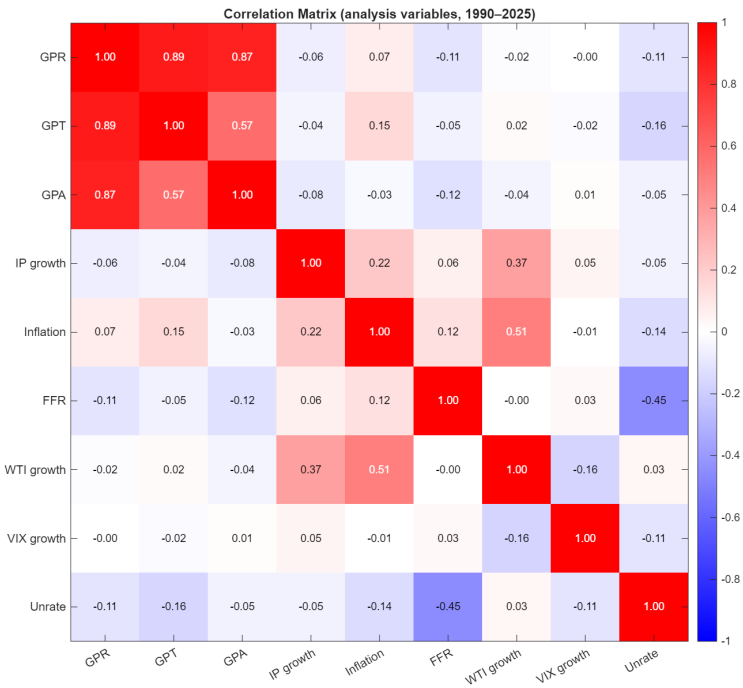

Figure 3. Correlation matrix of analysis variables, 1990–2025.

5.2 Persistence of the Shock Series

Figures 4 and 5 show the ACF and PACF of the three GPR variants. All three series exhibit strong persistence (ACF decays slowly), motivating the inclusion of shock lags in both ADL and LP specifications. PACF cuts off after a small number of lags, consistent with a low-order autoregressive representation.

5.3 ADL Impulse Responses

Figure 6 presents the baseline ADL IRFs (no controls; level response, 1-s.d. shock, 90% Monte Carlo CI).

IP responses. All three shock variants produce a persistent negative IP response. GPR and GPA generate statistically significant contractions at medium horizons ($h = 6$–24); GPT is qualitatively

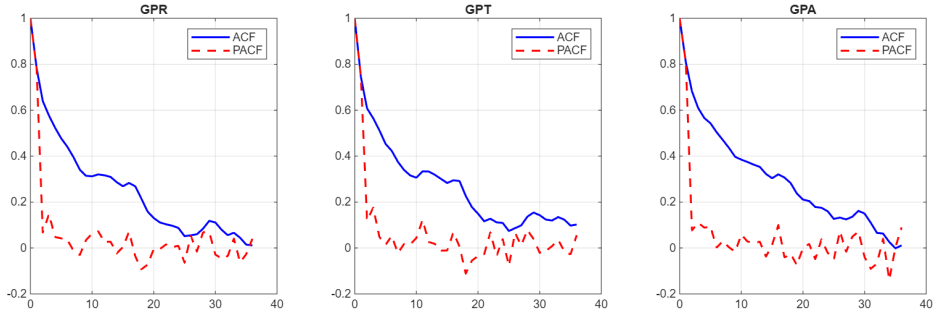

Figure 4. ACF (blue) and PACF (red dashed) of GPR / GPT / GPA, up to 36 lags.

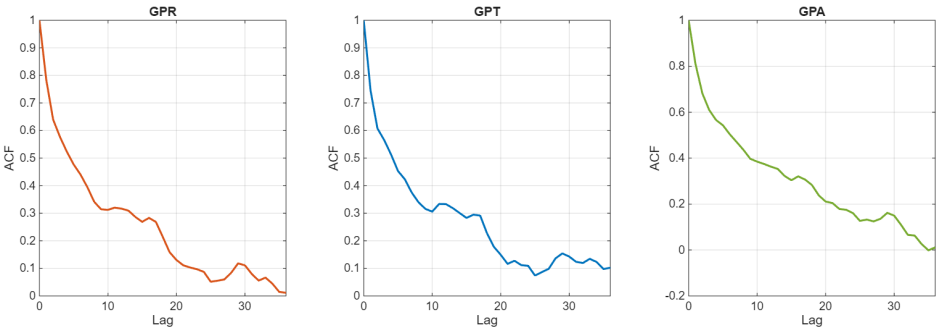

Figure 5. ACF of GPR / GPT / GPA (separate panels), up to 36 lags.

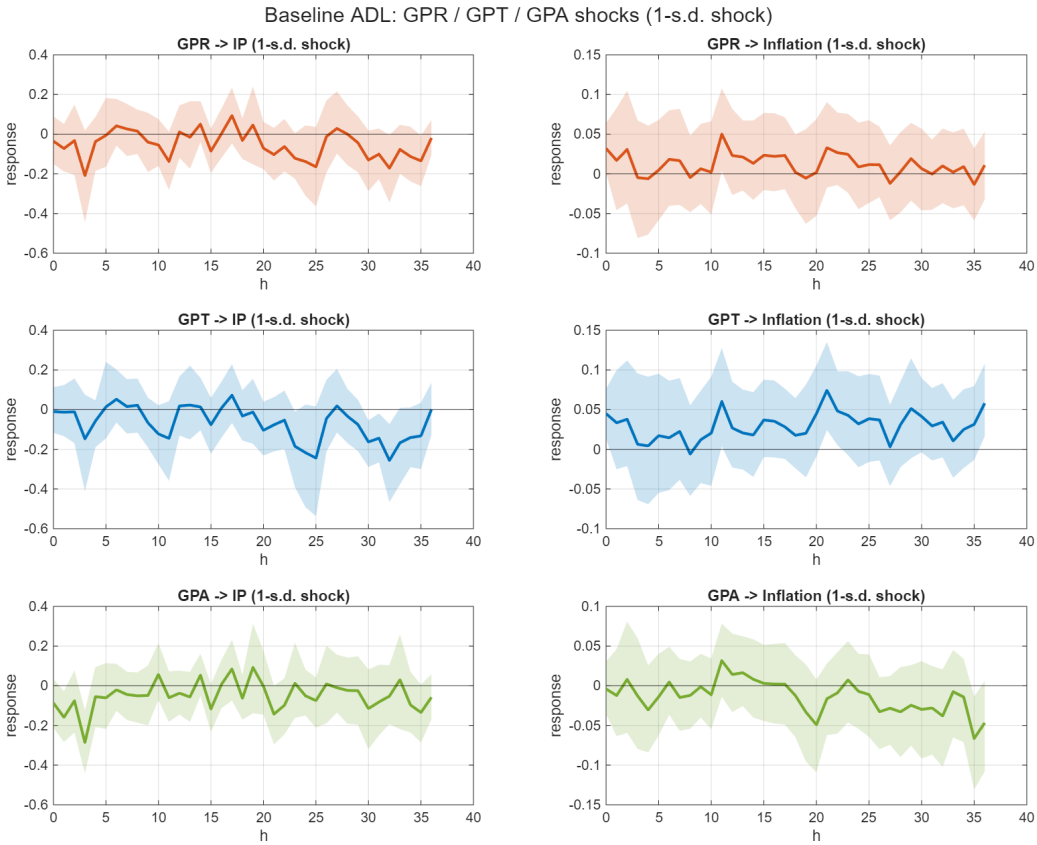

Figure 6. Baseline ADL: IP and CPI responses to GPR / GPT / GPA (1-s.d. shock, 90% CI).

similar but with wider confidence bands.

CPI responses. CPI responses are mixed and generally near zero, with confidence bands that include zero across all shock variants.

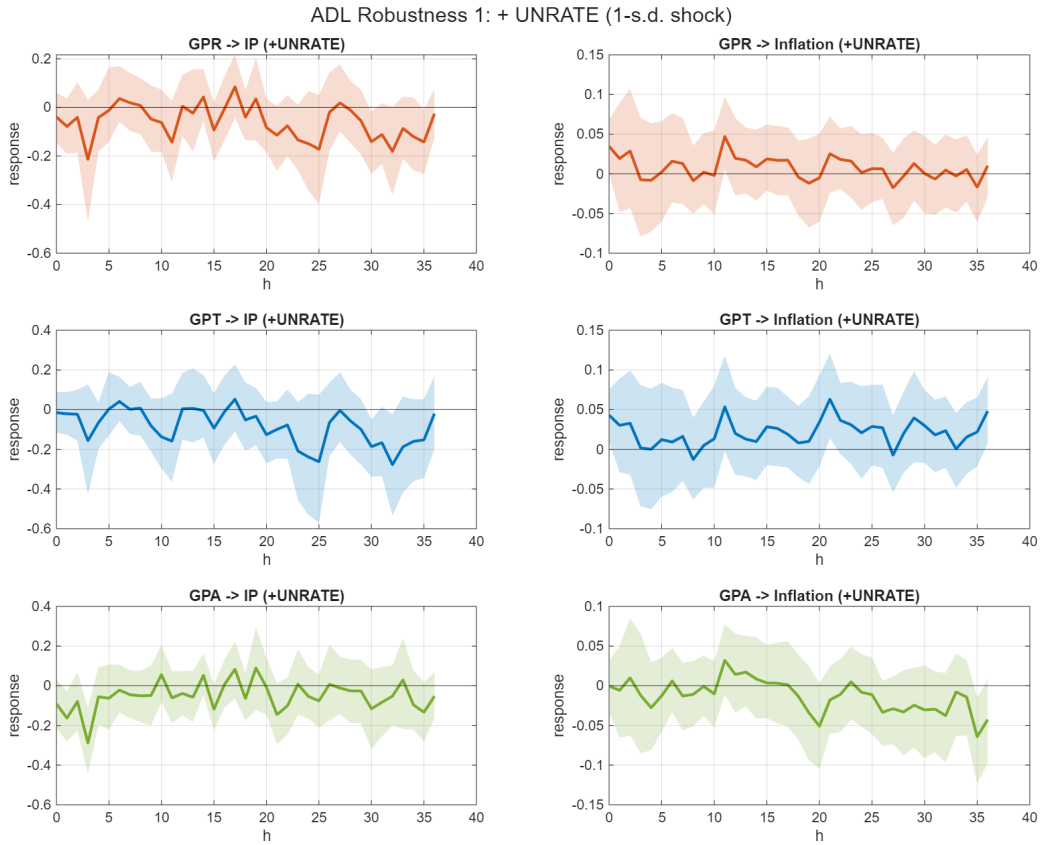

Figure 7. ADL Robustness 1: + UNRATE.

The specification-comparison overlay (Figure 9) shows that the ADL point estimates are largely unchanged by the addition of UNRATE and log(VIX), consistent with the GPR exogeneity assumption.

5.4 LP Impulse Responses

Figure 10 presents the baseline LP IRFs (own lags only, $L = 12$, 90% Newey–West HAC CI, 1-s.d. shock). Outcomes enter in log-level $\times 100$, so IRF units are log-points $\times 100$ ($\approx$ percent).

Result. The LP reproduces the ADL pattern qualitatively: a statistically significant IP contraction peaking at $h = 12$–24, and a CPI response that is near zero with confidence bands covering zero. The close alignment of the three specifications in Figure 13 confirms that the baseline LP estimates are not driven by business-cycle or financial-uncertainty confounding.

5.5 ADL vs. LP Cross-Check

Figures 14 and 15 overlay the baseline ADL and LP point estimates. The two methods agree on the sign, timing, and approximate magnitude of the IP contraction, and on the near-zero CPI response. LP confidence bands (Figure 15) fully contain the ADL point estimate at most horizons.

Agreement between ADL (parametric, with Monte Carlo inference) and LP (semi-parametric, with HAC inference) is informative: since the two methods rest on different auxiliary assumptions (ADL on a finite-order AR representation; LP on direct multi-step regressions), agreement strengthens confidence that the impulse response is not an artefact of the functional form.

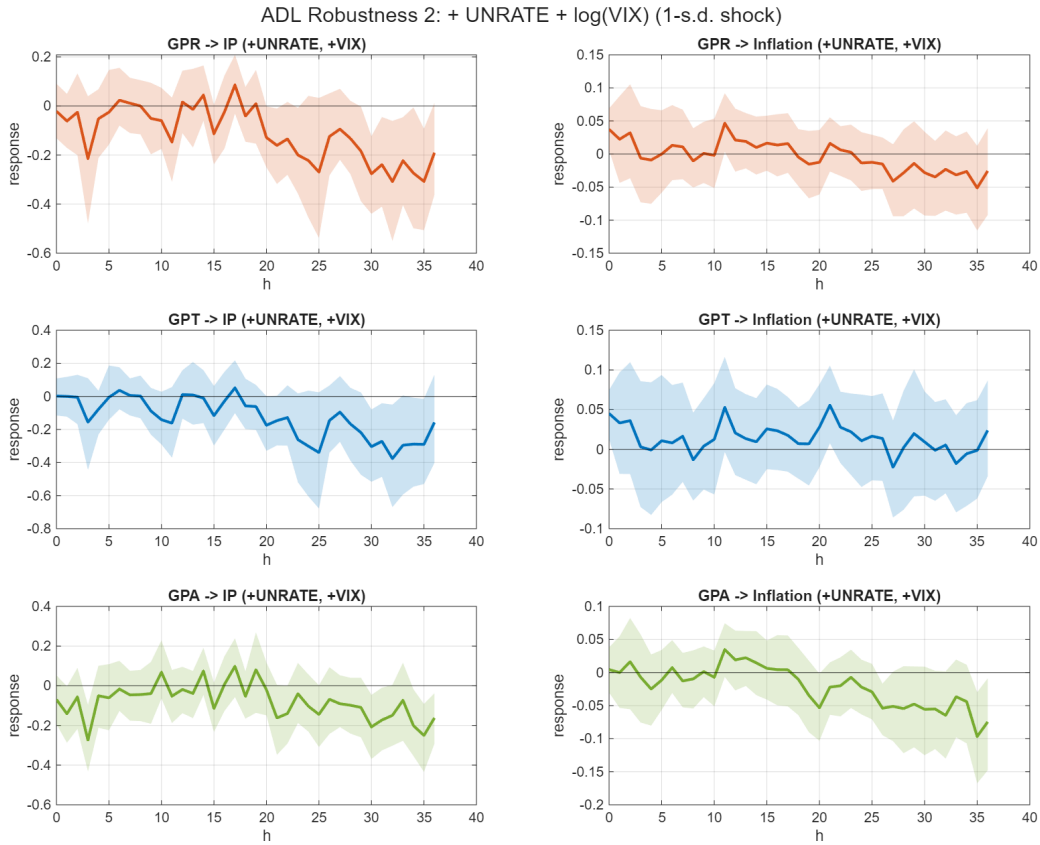

Figure 8. ADL Robustness 2: + UNRATE + log(VIX).

5.6 Summary of H1 Findings

Table 9. H1 Results Summary (baseline specification, 1-s.d. shock)

Shock	IP response	CPI response	Cost-push?
GPR (headline)	− (significant)	≈ 0 (insignificant)	Partial
GPT (threats)	− (marginal)	≈ 0 (insignificant)	Partial
GPA (acts)	− (significant)	≈ 0 (insignificant)	Partial

Conclusion. The *output contraction* half of the cost-push signature is confirmed by both methodologies: GPR shocks produce a statistically significant decline in industrial production at medium horizons, robust across specifications and methods. The *inflation* half (CPI ↑) is not significant in the reduced-form baseline.

5.7 Comparison with the Literature

Consistency with Caldara & Iacoviello (2022, AER). Our output result closely matches C&I's main finding: GPR shocks significantly reduce economic activity. C&I also report a weak price-level response in their baseline VAR and characterise GPR as closer to a *negative demand shock* than a pure cost-push shock in reduced form. Our LP- and ADL-based results are therefore fully consistent with their VAR evidence.

The CPI puzzle and Caldara et al. (2026, JIE). The title of Caldara et al. (2026) — *"Do Geopolitical Risks Raise or Lower Inflation?"* — itself signals that the inflation effect is theoretically and empirically ambiguous. Their key insight is that a GPR shock activates three opposing channels simultaneously:

(i) **Supply / cost-push:** GPR ↑ → oil & commodities ↑ → CPI ↑

(ii) **Demand:** GPR ↑ → uncertainty ↑ → consumption & investment ↓ → CPI ↓

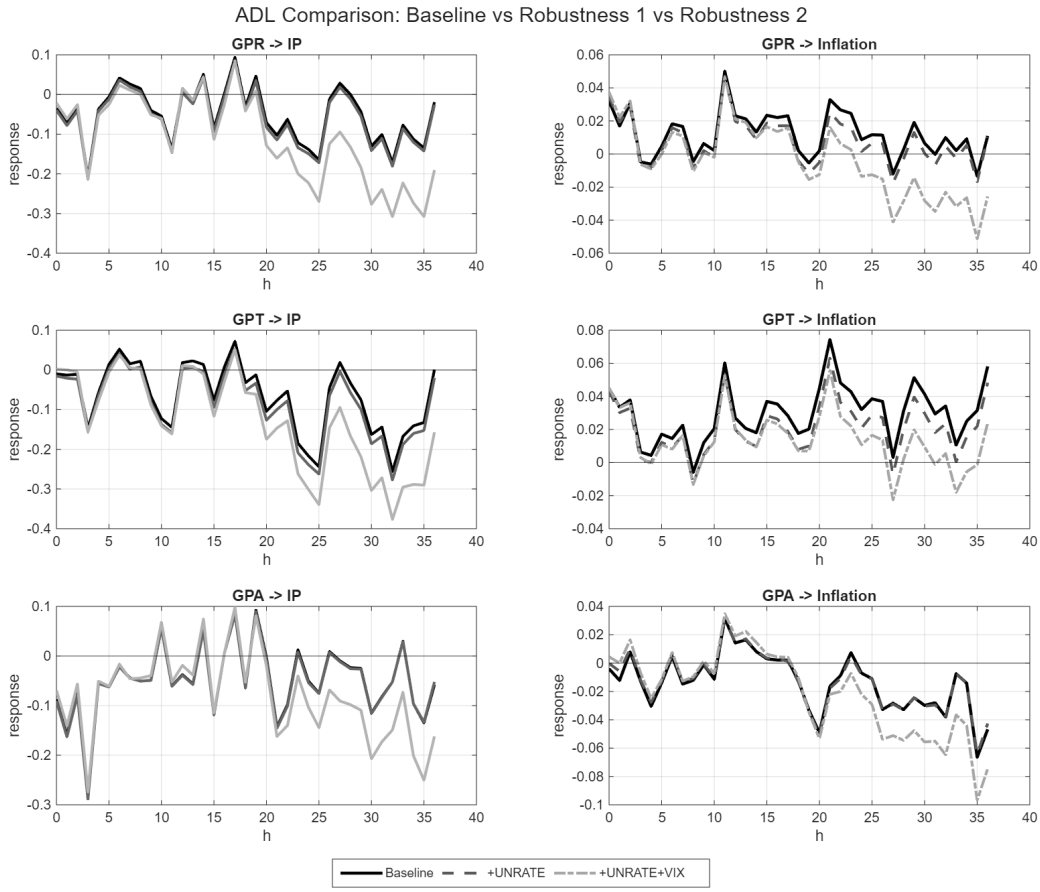

Figure 9. ADL specification comparison: Baseline vs. R1 vs. R2.

(iii) **Monetary-policy offset:** GPR $\uparrow \rightarrow \pi \uparrow \rightarrow$ Fed tightens $\rightarrow$ CPI $\downarrow$

Our reduced-form baseline captures the *net* effect. When supply-side inflationary pressure and demand-side/policy deflationary pressure roughly cancel, $\beta_h \approx 0$ is exactly the expected outcome. This is not a failure of the hypothesis — it is evidence of channel offset, which Caldara et al. (2026) document in their country-panel analysis.

GPT vs. GPA decomposition. GPA (geopolitical acts) shows a stronger and more persistent IP contraction than GPT (threats). This is consistent with C&I's (2022) interpretation: realised events disrupt supply chains and commodity markets more directly than forward-looking threat coverage, which operates primarily through expectations and financial-market risk premia.

Implications for Stage 4. The near-zero CPI response in the reduced-form LP/ADL strongly motivates the channel decomposition planned in Stage 4. Conditioning the LP on oil prices (WTI) as a mediator should:

- **Isolate the supply channel:** the LP of CPI on GPR *controlling for* WTI should show attenuation, confirming that the commodity-price channel is operative.
- **Reveal the demand channel:** the residual CPI response (net of oil) may turn negative, exposing the uncertainty-driven demand contraction.

Key methodological point. Whether a GPR shock is "cost-push" cannot be judged from the reduced-form net CPI effect alone. A reduced-form $\beta_h \approx 0$ is consistent with a strong supply-side inflationary pressure exactly offset by demand contraction and policy response. The correct test is whether the supply channel operates *independently*: if conditioning on oil prices attenuates the CPI response, the supply channel is confirmed as operative regardless of the net sign. This is the identification strategy for Stage 4.

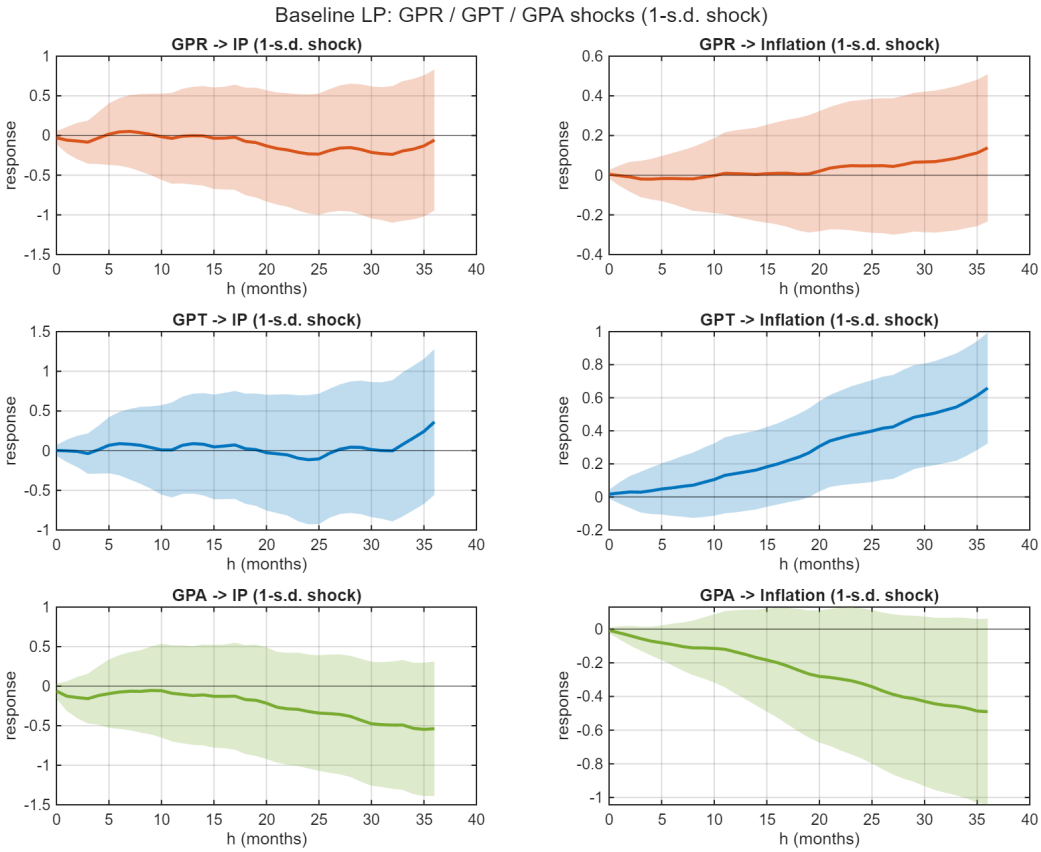

Figure 10. Baseline LP: IP and CPI level responses to GPR / GPT / GPA (1-s.d. shock, 90% CI).

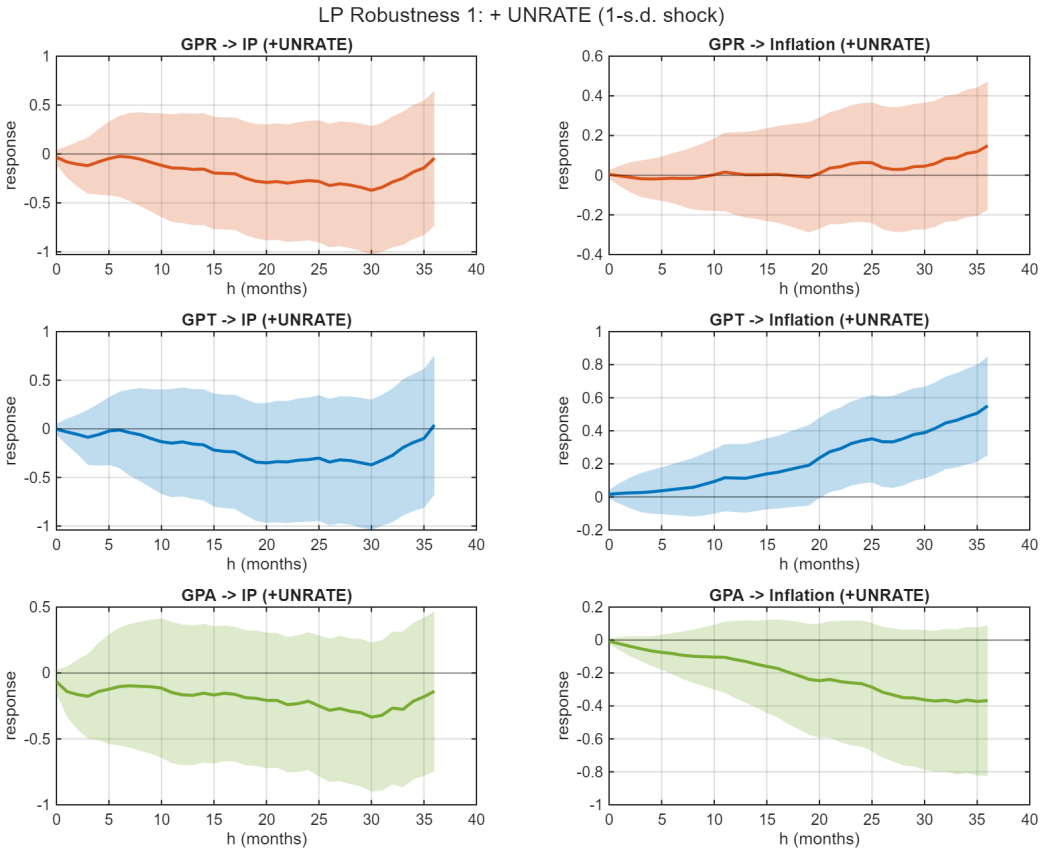

Figure 11. LP Robustness 1: + UNRATE.

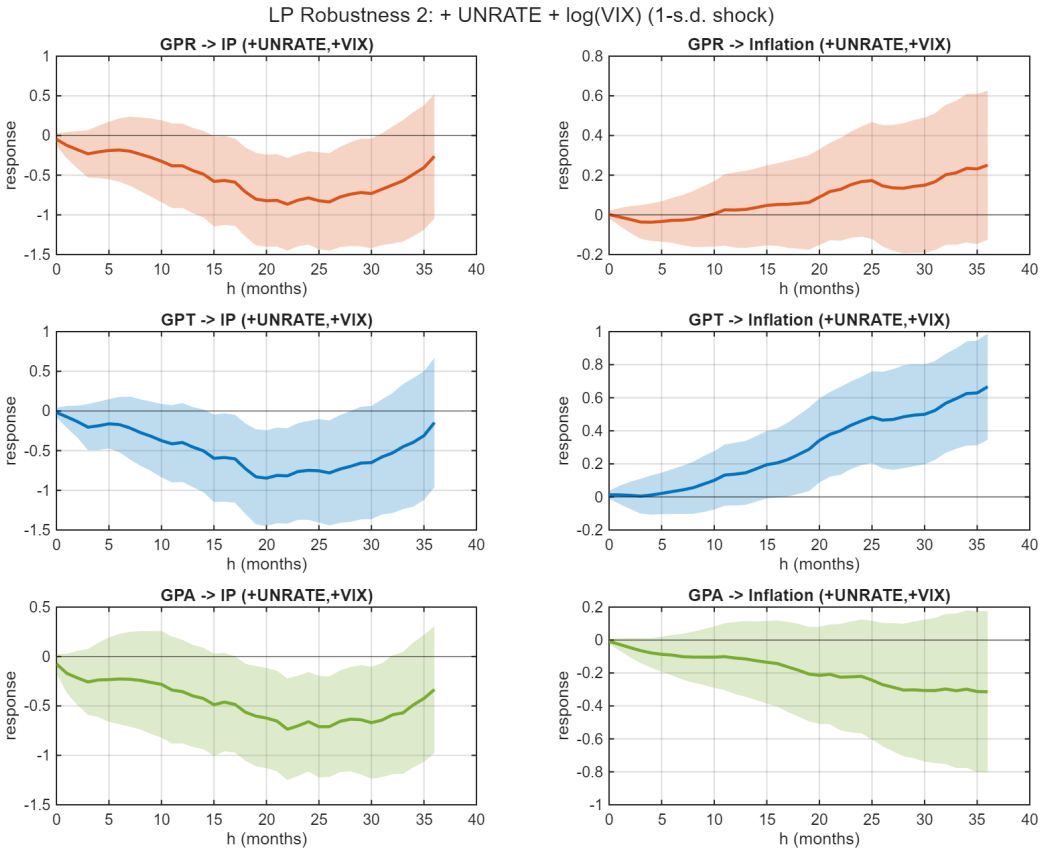

Figure 12. LP Robustness 2: + UNRATE + log(VIX).

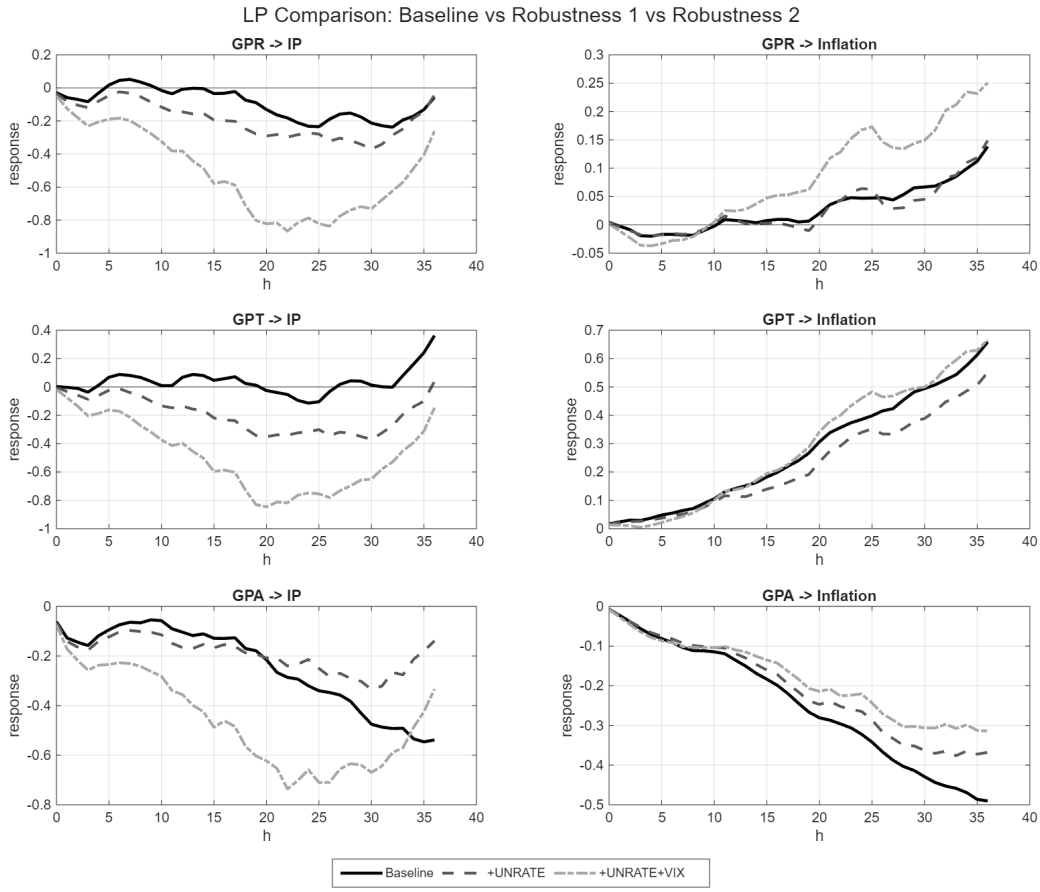

Figure 13. LP specification comparison: Baseline vs. R1 vs. R2.

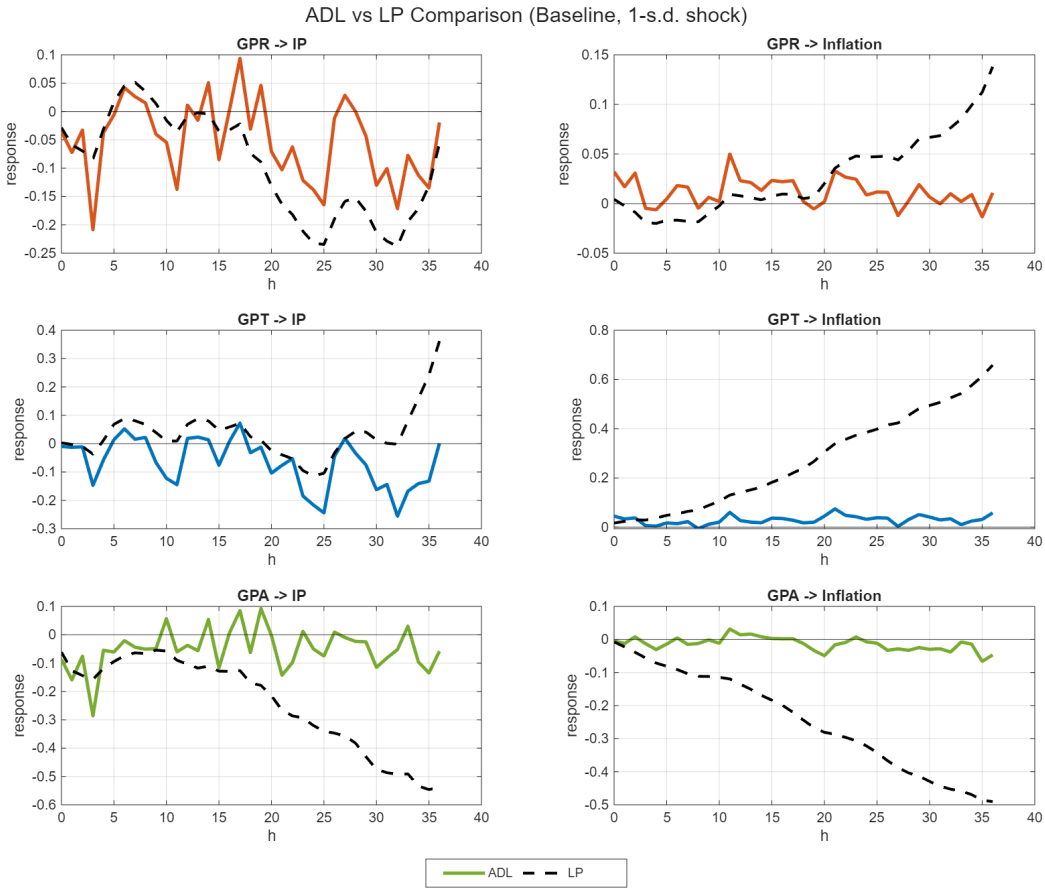

Figure 14. ADL vs. LP, baseline specification (1-s.d. shock, point estimates only).

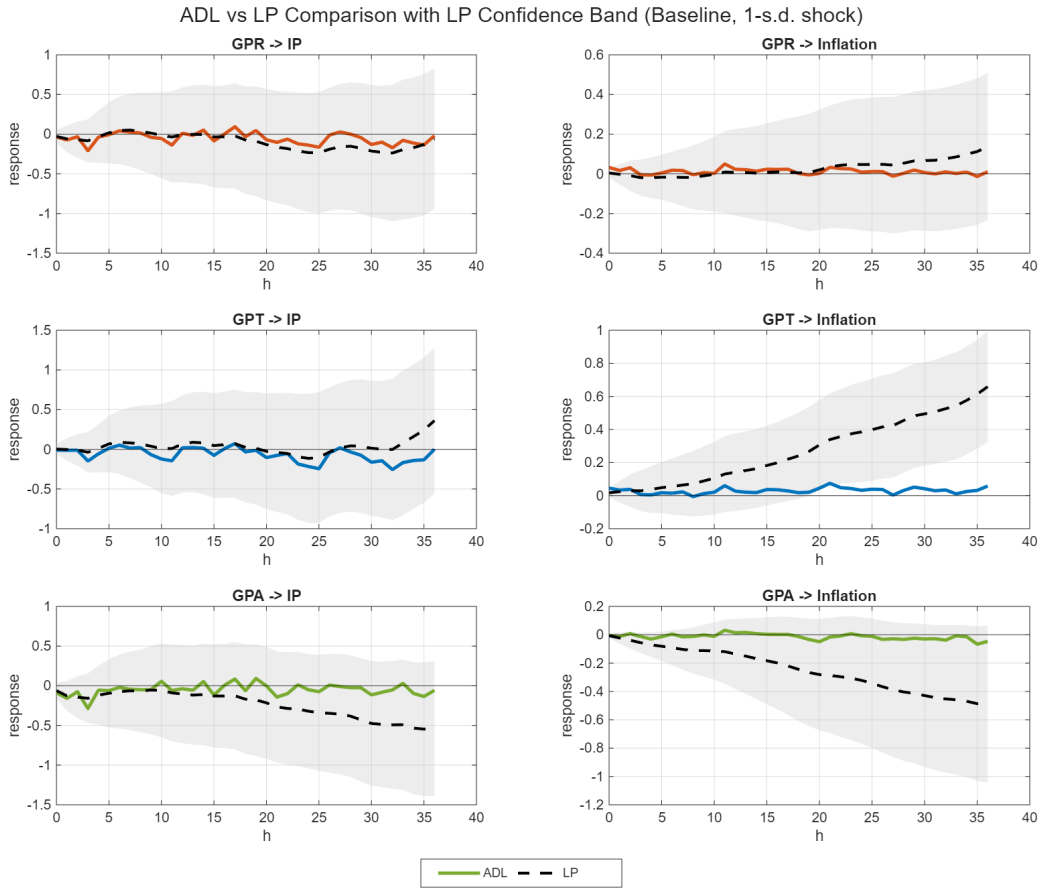

Figure 15. ADL vs. LP, baseline specification, with 90% LP confidence band.

6. Code Structure

6.1 Folder Layout

```
Stage1/
  run_h1_simple_v2.m        <- main analysis script (run this)
  h1_baseline.mat           <- pre-built panel (from Feedback_Pf.Baek/H1/run_all.m)
  ir_jorda.m, ir_ADL.m      <- LP and ADL estimators (Baek-Lee)
  simple_acf.m, simple_pacf.m <- ACF/PACF without Econometrics Toolbox
  lrv_nw.m, adltoma.m       <- HAC SE and ADL-to-MA conversion
  figures/                  <- all .png outputs produced by the script
```

6.2 How to Run

Prerequisites: MATLAB R2020a or later (base toolbox only; no Econometrics Toolbox required).

```
>> run('Stage1/run_h1_simple_v2.m')
```

The script trims the sample to 1990:01–2025:12, constructs all transformations, and produces the 15 figures listed in Table 10.

Table 10. Figure Inventory

File	Content	Section
fig01	GPR/GPT/GPA time series	Descriptive
fig02	Macro variables panel	Descriptive
fig03	Correlation matrix	Descriptive
fig04a,b	ACF and PACF	Persistence
fig05a–d	ADL: Baseline, R1, R2, comparison	ADL
fig06a–d	LP: Baseline, R1, R2, comparison	LP
fig07a,b	ADL vs LP overlay (point, point+CI)	Cross-check

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